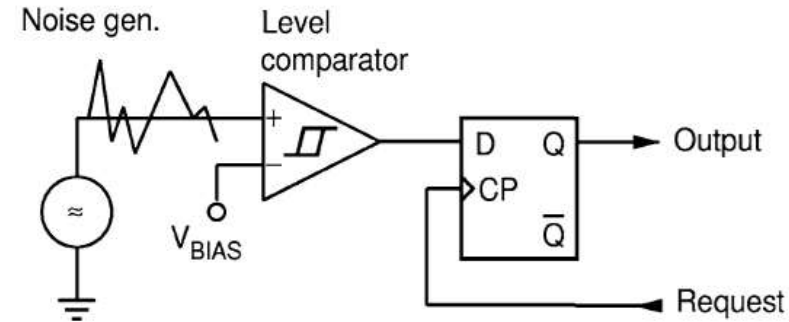
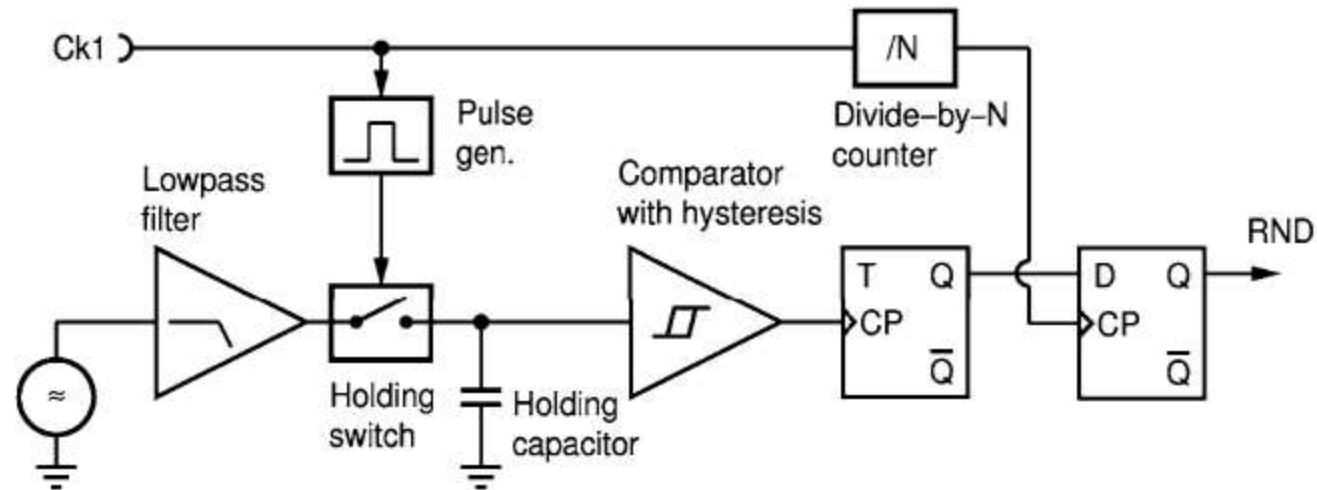


Noise-Based RNG



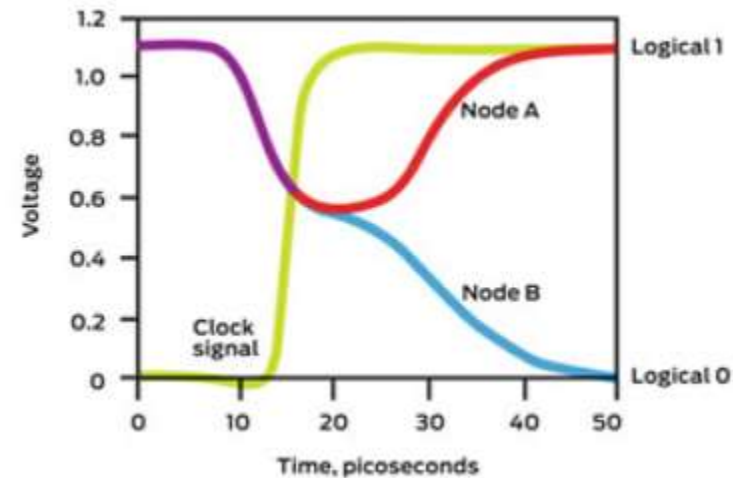
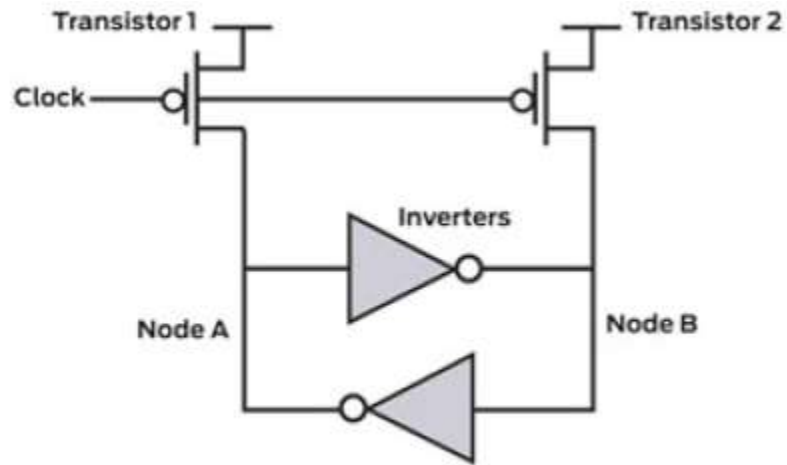
- Noise-based RNGs exploit Johnson's thermal noise in resistors or Zener noise in semiconductor diodes. Johnson's effect creates random voltage from thermal motion of charge carriers at any temperature above absolute zero.
- The fundamental challenge
 - Noise sources cannot be well characterized or controlled during fabrication.
 - Amplification introduces bandwidth limitations and nonlinearity.
 - Multiple nearby RNGs tend to synchronize through electromagnetic interference, dramatically reducing entropy.

Bagini-Bucci Generator



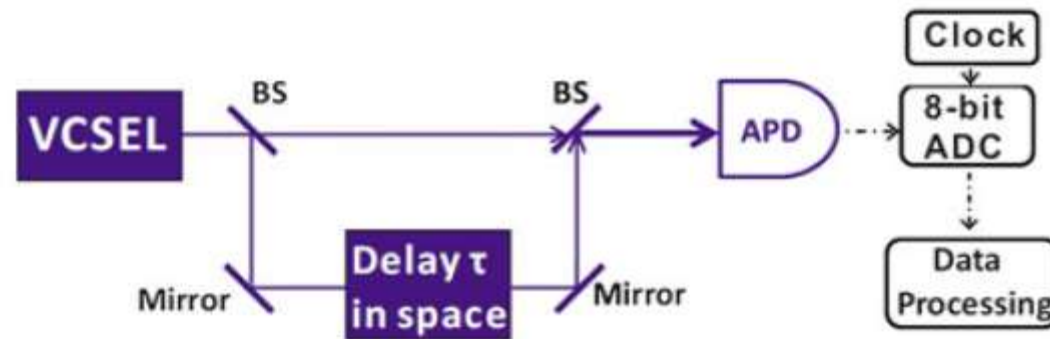
- Uses T flip-flop to divide biased output by 2, spending exactly 50% of time in each state.
- Requires sampling frequency much lower than noise frequency to approach randomness.

Intel's RNG



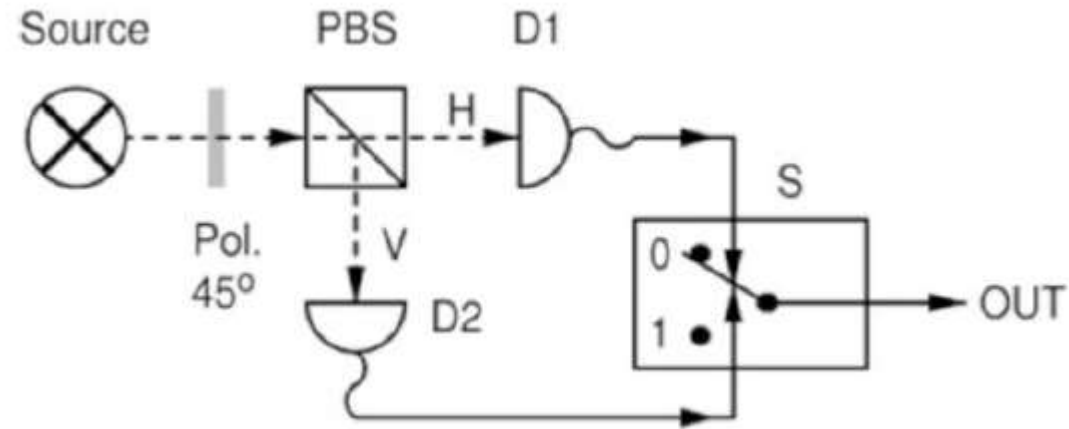
- The circuit has two stable states, but even smallest asymmetries cause high imbalance or complete lockup.

Laser Phase Noise Generators



- Single-mode VCSEL laser with signal and delayed copy brought to interference via Michelson-Morley interferometer.
- Phase jitter detected by high gain-bandwidth photodiode produces electrical noise.

Beam Splitter



- Circularly polarized photon enters polarizing beam splitter (PBS). PBS decomposes polarization, sending horizontal component to one port, vertical to another.
- Circularly polarized photon has equal content of both linear polarizations but cannot split in half—exactly 50% chance to exit either port. Label ports "0" and "1" for theoretically perfect RNG.